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{
  "creator" : "ari.neko@cvut.cz",
  "timestamp" : "10-07-2026_16:05:37",
  "variant" : 3,
  "subjectName" : "mathematics",
  "categories" : [ "Mathematical Logic", "Combinatorics", "Probability" ],
  "questions" : [ {
    "question" : "What is a logical statement called if it evaluates to true under all possible truth value assignments?",
    "answers" : [ "Contradiction", "Tautology", "Contingency", "Fallacy" ],
    "correctIndex" : 1
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    "question" : "What is the negation of the statement 'All primes are odd'?",
    "answers" : [ "There exists at least one prime that is not odd", "All primes are even", "No primes are odd", "Some primes are odd" ],
    "correctIndex" : 0
  }, {
    "question" : "Which principle is also known as the Dirichlet Box Principle?",
    "answers" : [ "Pigeonhole Principle", "Bijection Principle", "Multiplication Principle", "Inclusion-Exclusion Principle" ],
    "correctIndex" : 0
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    "question" : "How many distinct subsets can be formed from a set containing 5 elements?",
    "answers" : [ "25", "32", "16", "64" ],
    "correctIndex" : 1
  }, {
    "question" : "How many unique permutations can be made from the letters in the word 'APP'?",
    "answers" : [ "9", "3", "6", "1" ],
    "correctIndex" : 1
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    "question" : "In how many ways can 4 people be seated in a straight line?",
    "answers" : [ "6", "16", "12", "24" ],
    "correctIndex" : 3
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    "question" : "If events A and B are independent, what is  $P(A \text{ and } B)$ ?",
    "answers" : [ " $P(A) / P(B)$ ", " $P(A) + P(B)$ ", " $P(A) * P(B)$ ", " $P(A) + P(B) - P(A \text{ or } B)$ " ],
    "correctIndex" : 2
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    "question" : "What does Bayes' Theorem calculate?",
    "answers" : [ "Marginal probability", "Conditional probability", "Joint probability", "Discrete distributions" ],
    "correctIndex" : 1
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}, {  
  "question" : "What is the probability of rolling a sum of 7 with two fair six-sided dice?",  
  "answers" : [ "1/4", "1/12", "1/6", "1/36" ],  
  "correctIndex" : 2  
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  "question" : "What is the sum of probabilities of all possible mutually exclusive outcomes in a  
sample space?",  
  "answers" : [ "100", "Infinite", "0", "1" ],  
  "correctIndex" : 3  
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